

Automating User Authentication *with Authconfig*

Modern computer infrastructure tends to consist of many machines, with multiple services running on them. Keeping local user accounts in sync on all these machines and their services is a daunting task, even more so since passwords need to remain synced.



A solution to syncing multiple user accounts and passwords on a network of machines is not to store account information on local systems, but instead retrieve this information from a centralised store. Having user information and the associated authentication information centralised also allows for something called single sign on (SSO). With SSO, a user authenticates once using a password and then obtains a form of a ticket or cookie that can be used to automatically authenticate to other services.

User information and authentication

A centralised identity management system will need to provide at least two services.

Account information: This includes information such as user name, home directory location UID and GID, group memberships, etc. Popular solutions include LDAP (Lightweight Directory Access Protocol), which is used in multiple products such as Active Directory, IPA server and Network Information Services (NIS).

Authentication information: This is a way for a system to validate that users are who they claim to be. This can be done by providing a cryptographic password hash to the client system, or by sending the (encrypted) password to the server and receiving a response. An LDAP server can provide authentication information in addition to account information. Kerberos provides only SSO authentication services and is typically used alongside LDAP user information. Kerberos is used in both IPA server and Active Directory.

On a RHEL 7 system, local user information is provided by `/etc/passwd` while authentication information (in the form of a hashed password) is provided by `/etc/shadow`.

Attaching a system to centralised LDAP and Kerberos servers

Configuring a Red Hat Enterprise Linux 7 system to use centralised identity management services requires the editing of various files and the configuration of some daemons. For attaching to central LDAP and Kerberos server, the following